

EDUGENIE-AI

Coding, Solution & Functional Features

Team Members

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Project: EduGenie-AI

Code-Layout, Readability and Reusability

Project Name: EduGenie-AI

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Consistent indentation is used throughout the project.
2	Proper File Structure	Files and folders are logically organized	Yes	Frontend, backend, templates and static files are organized separately.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables and functions use descriptive names.
4	Function / Method Names	Functions are descriptively named	Yes	Function names describe the operation performed.
5	Code Comments	Inline and block comments explain logic	Yes	Comments are added where clarification is useful.
6	Modular Design	Code is split into reusable functional modules	Yes	Application logic is separated into reusable modules/services.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Repeated and unnecessary code is minimized.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	API and runtime errors are handled appropriately.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Frontend Interface	HTML / CSS / JavaScript	Student-facing EduGenie-AI interface	High
2	FastAPI Backend	Python / FastAPI	Request handling and API endpoints	High
3	AI Service	Google Gemini API	AI-generated educational responses	High
4	Configuration	Python / Environment Variables	Application and API configuration	High
5	Static Assets	CSS / JavaScript	Styling and browser-side interaction	Medium

Code-Layout, Readability and Reusability

Reusable Components / Modules — Continued

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Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clear project structure with separate application modules.
Readability	5	Naming and organization make the code easy to follow.
Reusability	5	Services, configuration and interface components can be reused.
Documentation / Comments	5	Important logic and project usage are explained clearly.
Overall Score	5	EduGenie-AI follows a structured and maintainable implementation.

Coding & Solution

Project Name: EduGenie-AI

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g. GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	GitHub repository for the EduGenie-AI project
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	FastAPI
Key Features Implemented	AI-powered educational assistance; student query processing; Gemini API integration; web-based interface; structured backend API
Pending / Incomplete Features	Optional enhancements such as additional learning modules and extended personalization
Setup / Run Instructions	Create and activate a Python virtual environment, install dependencies, configure the Gemini API key in the environment file, and run the FastAPI application with Uvicorn.

Coding & Solution

Code Quality Checklist

S.N	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

EduGenie-AI is organized as a web-based AI application. The frontend collects the learner's request, the FastAPI backend processes it, and the Gemini AI service generates the educational response. The project is structured for maintainability and future feature expansion.

No. of Functional Features Included in the Solution

Project Name: EduGenie-AI

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)
1	Student Query Input	Allows students to enter educational questions or learning requests.	Frontend / UI	Done
2	AI Response Generation	Generates relevant educational responses using generative AI.	Gemini AI Service	Done
3	Natural Language Interaction	Accepts questions in natural language for easier student interaction.	Frontend + AI Service	Done
4	API Request Processing	Receives and processes requests between the interface and AI service.	FastAPI Backend	Done
5	Response Display	Presents the generated answer clearly to the student.	Frontend / UI	Done
6	Configuration Management	Keeps application and API configuration organized through environment settings.	Configuration Module	Done
7	Error Handling	Handles API and application errors and provides an appropriate response.	Backend	Done
8	Web-Based Access	Provides EduGenie-AI through a browser-based application interface.	Frontend + Backend	Done

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

No. of Functional Features Included in the Solution

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Question input, response display, browser interface
2	Backend / Logic	2	Request processing, error handling
3	Database / Storage	0	No database is required for the core implementation
4	API / Integration	2	FastAPI endpoints, Gemini API integration
5	Security / Authentication	1	Environment-based API key configuration

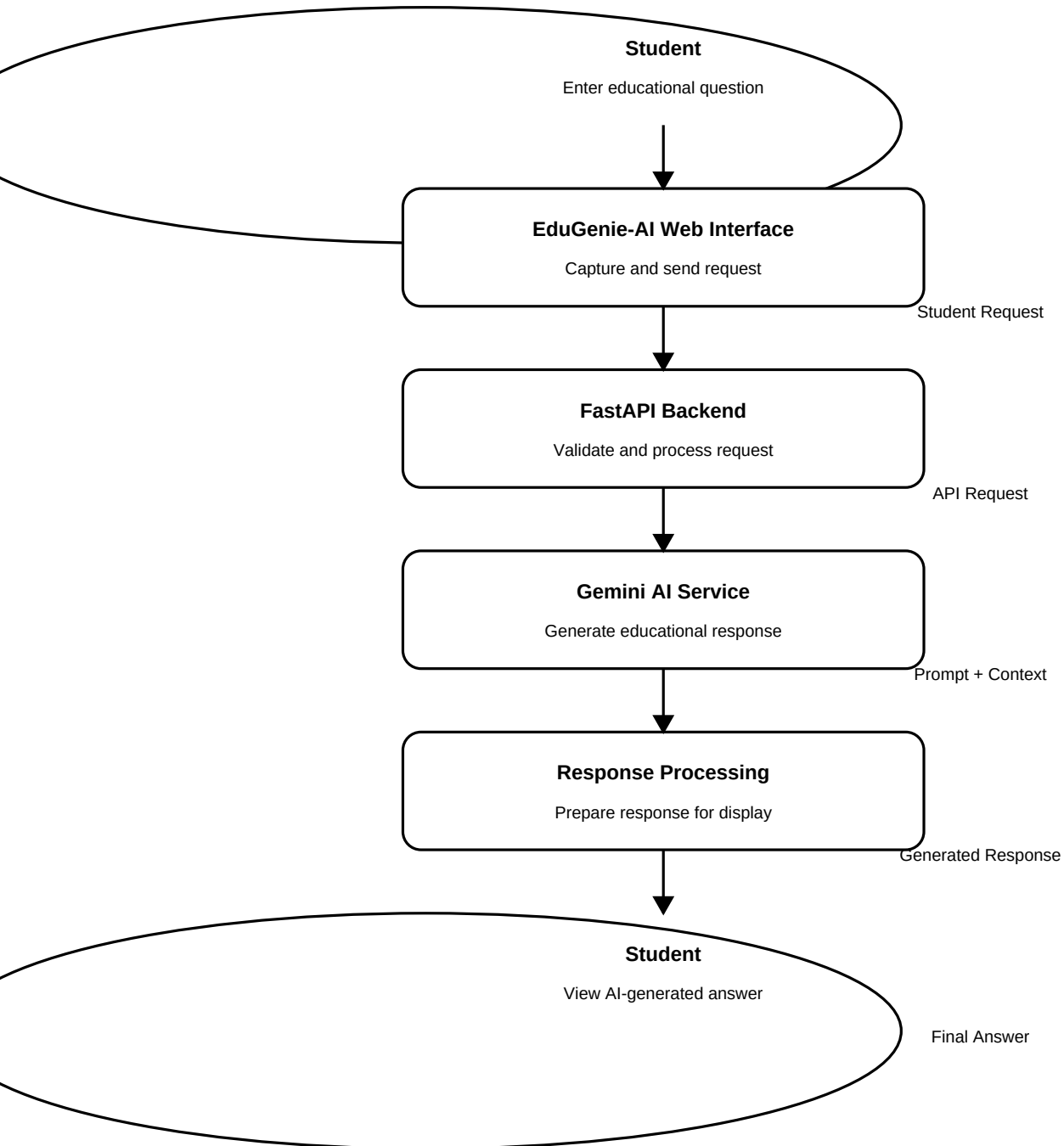
EduGenie-AI Feature Summary

EduGenie-AI combines a browser-based interface, a FastAPI backend, and a Gemini-powered AI service. The main workflow is designed to accept a student's learning request, process it through the backend, generate an educational response, and display the result clearly.

EduGenie-AI Flow Diagram

System Flow

The following flow shows the complete movement of a student's request through EduGenie-AI and back to the student.



Flow: Student → EduGenie-AI Web Interface → FastAPI Backend → Gemini AI Service → Response Processing → Student